St. John College of Engineering and Management
Autonomous Institute
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DTE Code : 3218 AICTE Permanent ID : 1-4790201
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Capstone Project Report on

PLC-Controlled Automatic Parking Space Signalling System

Submitted in partial fulfillment of the requirements of the Third Year of

Bachelor of Technology in

Electronics and Computer Science

by

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DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI

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Certificate of Approval

This is to certify that, following students of T. Tech **in Electronics and Computer Science Department**. They have satisfactorily completed the requirements for the courses Programmable Logic Controllers and SCADA prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT** (An Autonomous College affiliated to University of Mumbai), while working on "PLC-Controlled Automatic Parking Space Signalling System".

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Declaration

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Abstract

The PLC-Controlled Automatic Parking Space Signalling System is designed to monitor parking space availability and provide clear visual indications to drivers. The system uses a Programmable Logic Controller (PLC) to process signals from sensors installed in individual parking slots. The main objective of this project is to reduce the time required to find an available parking space and improve the efficiency of parking management.

In this system, three input sensors, X0, X1, and X2, are used to detect the occupancy status of three parking slots. The PLC processes these inputs according to the ladder logic program. When all three parking slots are occupied, the internal relay M0 is activated, indicating that the parking area is full. The output Y1 operates a red LED to indicate that no parking space is available. When at least one slot is vacant, the relay remains inactive, and output Y0 activates a green LED to indicate that parking space is available. Additionally, outputs Y2, Y3, and Y4 can be used as optional indicators to show the availability of individual parking slots.

The proposed system provides a simple, reliable, and cost-effective solution for parking space monitoring. It reduces manual supervision, improves convenience for drivers, and demonstrates the practical application of PLC technology in automated parking management systems.

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CHAPTER 1: OVERVIEW

1.1 Introduction

Parking management is an important concern in places such as shopping malls, hospitals, and educational institutions. Drivers often waste time searching for vacant parking spaces, causing traffic congestion and fuel wastage. An automatic parking space signalling system helps solve this problem by indicating parking availability.The PLC-Controlled Automatic Parking Space Signalling System uses a Programmable Logic Controller (PLC) to monitor three parking slots through sensors X0, X1, and X2. The PLC processes these inputs using a ladder logic program and controls the output indicators. The internal relay M0 activates when all three slots are occupied. A green LED (Y0) indicates available parking space, while a red LED (Y1) indicates that the parking area is full. Outputs Y2, Y3, and Y4 can be used for individual slot indicators.This system reduces manual supervision, saves time, and improves the efficiency of parking management.

1.2 Background

Traditional parking systems often require manual observation to identify vacant parking spaces, which can be time-consuming and inefficient. To overcome this problem, automation can be used to monitor parking availability. In this project, a Programmable Logic Controller (PLC) monitors three parking slots using input sensors X0, X1, and X2. The ladder logic program processes these inputs and controls LED indicators to display parking status. A green LED indicates that parking space is available, while a red LED indicates that all slots are occupied. This project demonstrates the use of PLC technology for simple and efficient parking management.

1.3 Importance of the Project

The PLC-Controlled Automatic Parking Space Signalling System is important because it helps drivers identify parking availability quickly and easily. It reduces the time spent searching for vacant spaces, minimizes unnecessary vehicle movement, and helps reduce traffic congestion. By using a PLC, sensors, and LED indicators, the system provides a simple and reliable method of monitoring parking slots. It also reduces the need for manual supervision and demonstrates the practical application of industrial automation in parking management. Furthermore, the system helps save fuel by reducing the time vehicles spend searching for parking spaces. It can also be

expanded in the future to manage a larger number of parking slots, making it useful for shopping malls, hospitals, offices, and other public places.

1.4 Perspective of Stakeholders and Customers

The main objective of the PLC-Controlled Automatic Parking Space Signalling System is to monitor parking slot occupancy and provide clear indications of parking availability. The system uses sensors X0, X1, and X2 to detect the status of three parking slots and a PLC to process the input signals through ladder logic. A green LED indicates that parking space is available, while a red LED indicates that all three slots are occupied. The project also aims to reduce the time required to find vacant parking spaces, minimize manual supervision, and improve the efficiency of parking management. Another objective is to demonstrate the practical application of PLC programming and automation in a real-world system. The design can also serve as a foundation for future improvements, such as expanding the number of parking slots and introducing advanced vehicle detection methods.

1.5 Objectives and Scope of the Project

Objectives:

The main objective of this project is to develop a PLC-controlled automatic parking space signalling system that monitors three parking slots and indicates their availability using LED indicators. The system uses sensors X0, X1, and X2 to detect parking occupancy, while the PLC processes the signals through ladder logic. A green LED indicates that parking space is available, whereas a red LED indicates that all three slots are occupied. The scope of this project includes reducing manual supervision, saving time for drivers, and improving parking management. The system can also be expanded in the future to monitor more parking slots and support advanced parking automation features.

Scope:

The scope of this project is to develop a PLC-based system that monitors three parking slots and indicates parking availability using LED indicators. It uses sensors, PLC ladder logic, and output signals to identify whether parking spaces are vacant or occupied. The system can be implemented in small parking areas and expanded in the future to support more parking slots and advanced parking management features.

CHAPTER 2: LITERATURE SURVEY & PROPOSED WORK

2.1 Introduction

This chapter discusses the existing parking management systems and the proposed PLC-Controlled Automatic Parking Space Signalling System. Traditional parking methods often require manual monitoring, which can be time-consuming and inefficient. The proposed system uses a Programmable Logic Controller (PLC), input sensors, and LED indicators to monitor three parking slots and display their availability. This chapter also explains the problems associated with existing methods, the proposed solution, and the feasibility of implementing the system using PLC ladder logic.

2.2 Existing Parking Management System

Traditional parking management systems mainly depend on manual observation to identify vacant parking spaces. In some modern systems, sensors, cameras, and digital displays are used to monitor parking availability. However, these systems may involve higher installation costs and complex maintenance. Manual methods can also waste time and cause unnecessary vehicle movement. The PLC-Controlled Automatic Parking Space Signalling System provides a simple alternative by using sensors, PLC ladder logic, and LED indicators to display parking availability efficiently and reduce the need for manual supervision.

2.3 Literature Survey

Table 1: Literature Survey of Parking Management Approaches

S. No.	System / Approach	Description	Limitations
1	Traditional Parking System	Uses manual observation to identify vacant parking spaces.	Time-consuming and requires manual supervision.
2	Sensor-Based Parking System	Uses sensors to detect the occupancy of parking slots.	Requires proper sensor installation and maintenance.
3	Camera-Based Parking System	Uses cameras to monitor parking areas and identify vacant spaces.	Can involve higher installation costs and complex processing.
4	PLC-Based Parking System	Uses a PLC to process sensor inputs and control parking indicators.	Requires PLC hardware and basic programming knowledge.
5	Proposed System	Uses three input sensors, PLC ladder logic, and LED indicators to display parking availability.	Currently designed to monitor only three parking slots.

2.4 Feasibility Study

The proposed PLC-Controlled Automatic Parking Space Signalling System is feasible in terms of technical, economic, and operational aspects. Technical feasibility: The system uses a PLC, three input sensors, LED indicators, and ladder logic, which are suitable for implementing a basic parking monitoring system. Economic feasibility: The project can be developed using a limited number of components, making it a cost-effective solution for small parking areas. Operational feasibility: The system is simple to operate and reduces the need for manual supervision by displaying parking availability through green and red LEDs. Therefore, the proposed system is practical for demonstrating automatic parking monitoring and can be expanded in the future to support additional parking slots.

2.5 Methodology Used

The methodology of this project involves designing and implementing a PLC-based automatic parking space signalling system. First, three input sensors, X0, X1, and X2, are used to detect the occupancy status of the parking slots. These inputs are connected to the PLC, which processes the signals according to the programmed ladder logic. The internal relay M0 is activated when all three

parking slots are occupied. Based on the PLC output, a green LED (Y0) indicates that parking space is available, while a red LED (Y1) indicates that all slots are occupied. Finally, the ladder logic is tested to verify the correct operation of the sensors and indicators under different parking condition.

CHAPTER 3: ANALYSIS AND PLANNING

3.1 Introduction

This chapter focuses on the analysis and planning of the PLC-Controlled Automatic Parking Space Signalling System. It describes the system requirements, hardware components, software tools, and PLC input-output configuration. The system uses three input sensors to detect parking occupancy, while the PLC processes the signals through ladder logic to control the LED indicators. Proper planning and analysis help ensure that the system operates efficiently, reliably, and according to the project requirements.

3.2 Project Planning

Table 2: Project Planning & Phase Breakdown

Phase	Activity Description	Deliverable	Duration
Phase 1	Requirement Analysis & Literature Review	Problem Statement & Component Selection	Weeks 1–2
Phase 2	CAD Design of Hopper, Adjustable Funnel & Rotating Disk	3D Printed Dispenser Assembly	Weeks 3–4
Phase 3	PLC I/O Mapping & Ladder Logic Development in GX Works3	Simulated & Verified .gx3 Ladder Program	Weeks 5–7
Phase 4	ESP32 Firmware, OLED Integration & Firebase Database Setup	Bi-directional Wi-Fi Cloud Bridge	Weeks 8–9
Phase 5	Android App Development (UI, OCR Scanner, AI Chatbot)	Functional APK Connected to Firebase	Weeks 10–11
Phase 6	Full System Integration, Fault-Injection Testing & Report	Final Prototype & Capstone Report	Week 12

CHAPTER 4: DESIGN & IMPLEMENTATION

4.1 Data Flow Diagram (DFD)

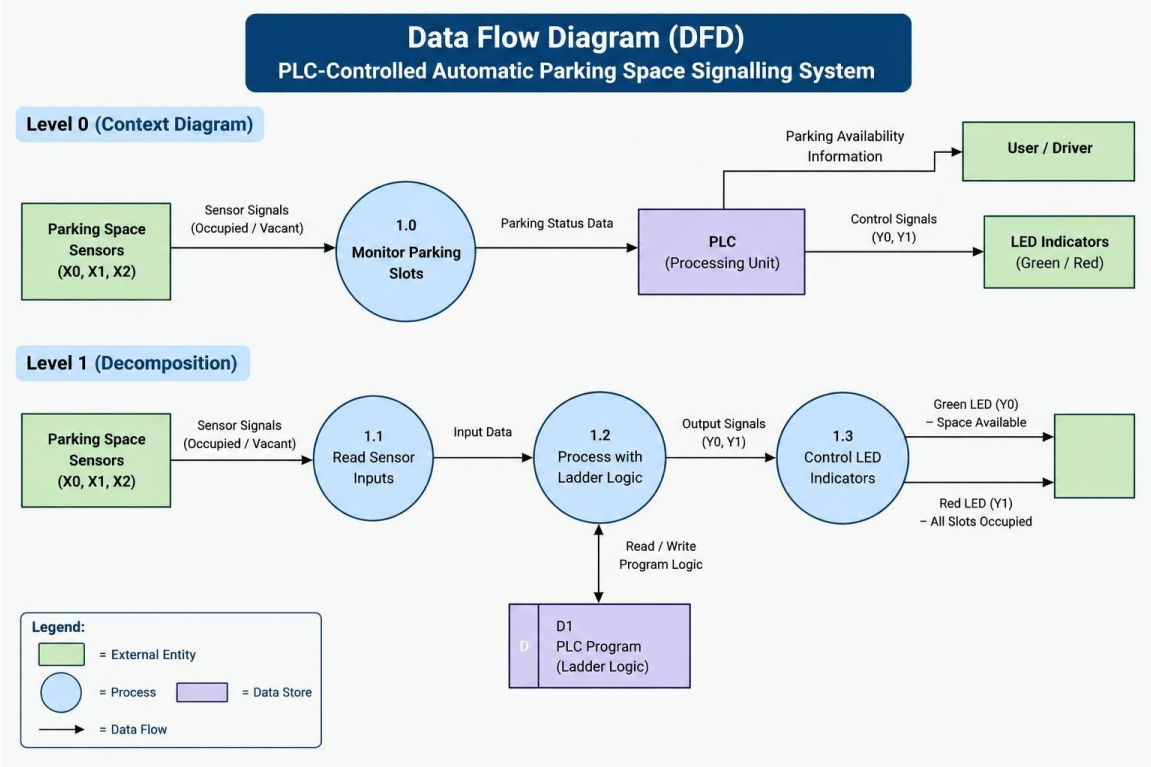


Figure 1: DFD Level 0 & Level 1

4.2 Block Diagram

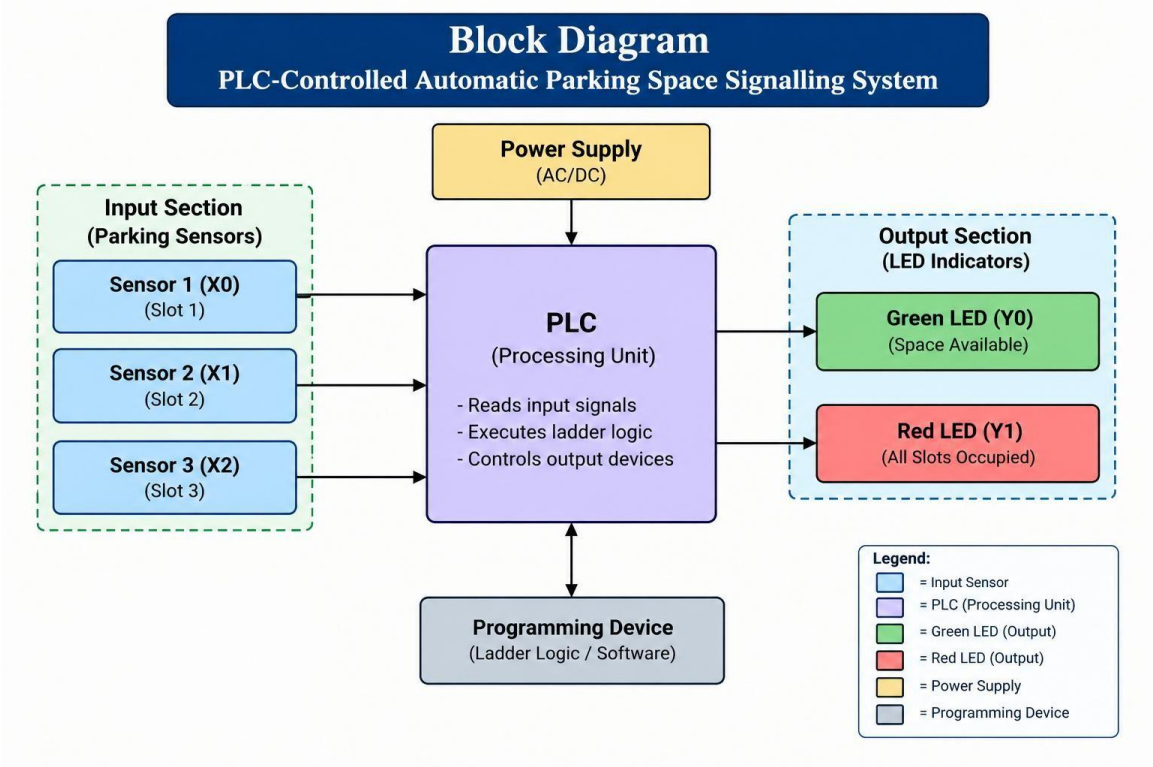


Figure 2: System Hardware & PLC Interfacing Block Diagram

4.3 Flow Chart

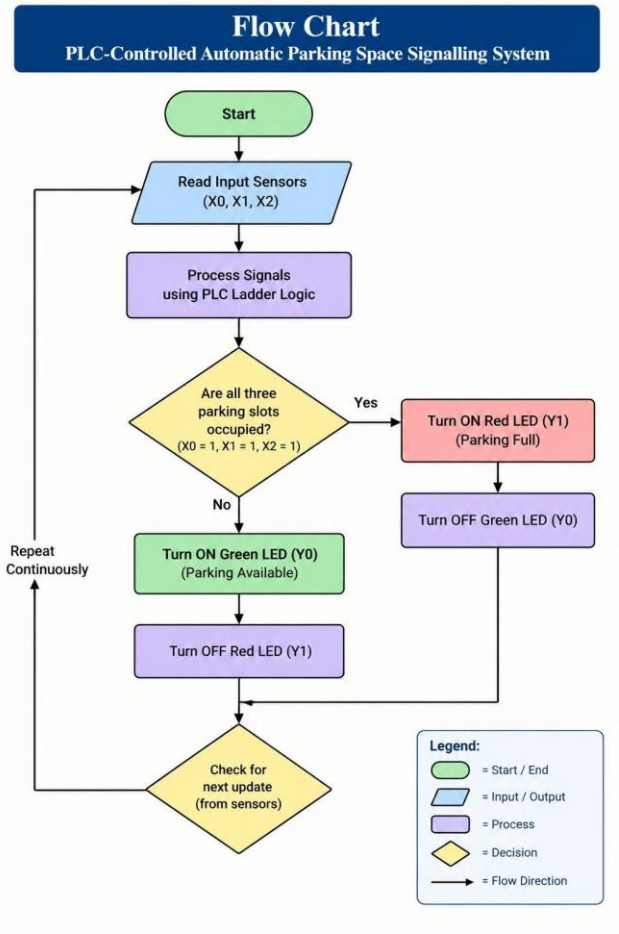


Figure 3: Dispensing & Fault Watchdog Sequence Flow Chart

4.4 UML & PLC Ladder Diagram

Table 3: Mitsubishi FX5U PLC I/O & Internal Relay Allocation

S. No.	Activity	Description
1	Start	The system begins operation.
2	Read Sensors	Sensors X0, X1, and X2 detect the occupancy status of the three parking slots.
3	Process Inputs	The PLC processes the sensor signals using ladder logic.
4	Check Parking Status	The system checks whether all three parking slots are occupied.
5	Parking Full	If all slots are occupied, the red LED Y1 turns ON and the green LED Y0 turns OFF.
6	Space Available	If at least one slot is vacant, the green LED Y0 turns ON and the red LED Y1 turns OFF.
7	Update Status	The PLC continuously checks the sensors and updates the indicators.

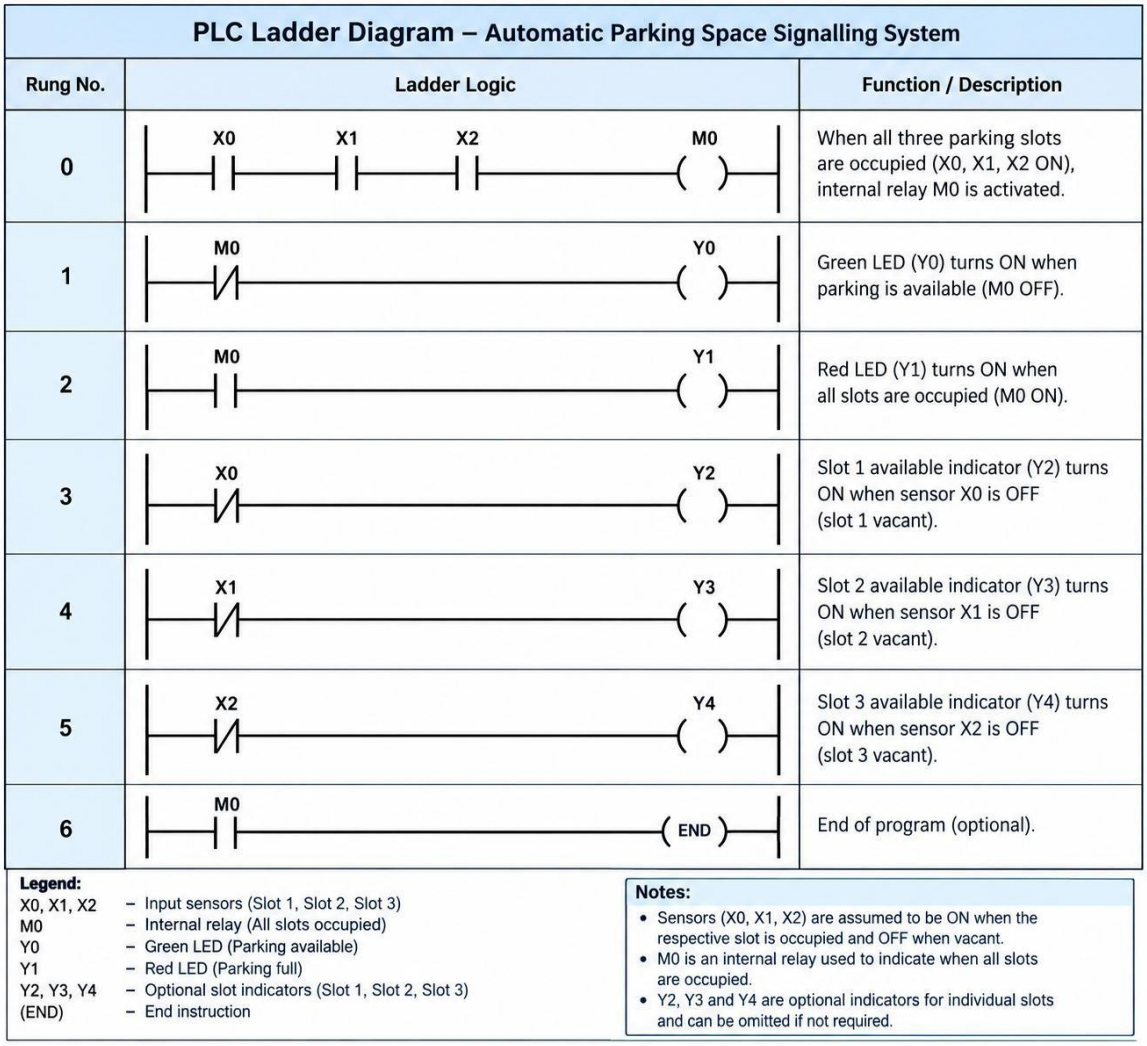


Figure 4: Mitsubishi FX5U PLC Ladder Diagram (MELSOFT GX Works3 Format)

CHAPTER 5: RESULTS & DISCUSSION

5.1 Actual Results

The PLC-Controlled Automatic Parking Space Signalling System was designed to monitor three parking slots using input sensors X0, X1, and X2. The ladder logic processes the sensor signals and controls the LED indicators according to the parking status. When at least one parking slot is vacant, the green LED (Y0) indicates that parking space is available. When all three slots are occupied, the internal relay M0 activates, turning ON the red LED (Y1) to indicate that the parking area is full. The optional outputs Y2, Y3, and Y4 can be used to indicate the status of individual parking slots. The system demonstrates how PLC-based automation can simplify parking monitoring, reduce manual supervision, and provide clear information about parking availability.

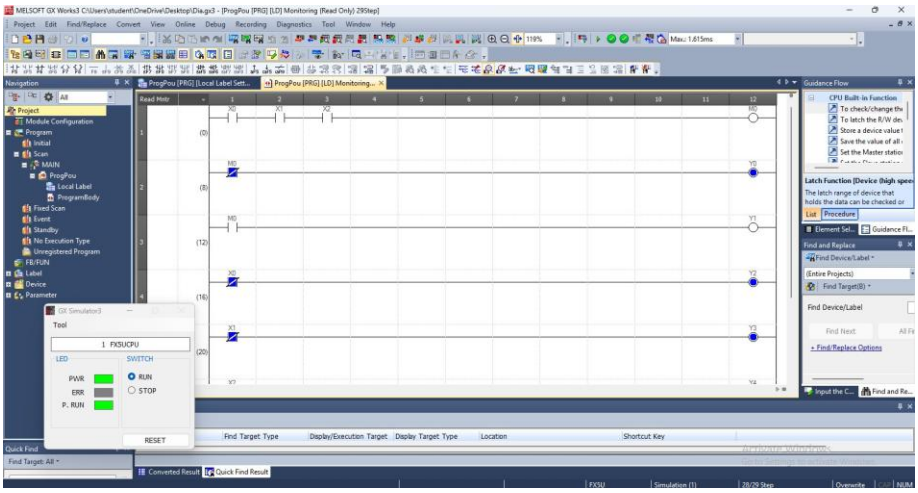


Figure 5: PLC Ladder Program Simulation in GX Works3

5.2 Future Scope

The PLC-Controlled Automatic Parking Space Signalling System can be improved in the future by increasing the number of parking slots monitored by the PLC. An LCD display can be added to show the number of available spaces, and IoT technology can be integrated to provide real-time parking information through a mobile application. Automatic entry and exit barriers can also be introduced to improve vehicle management.

Furthermore, advanced sensors can enhance vehicle detection and system reliability. These improvements can make the system more efficient, user-friendly, and suitable for large parking areas such as shopping malls, hospitals, and commercial buildings.

CHAPTER 6: CONCLUSION

The PLC-Controlled Automatic Parking Space Signalling System provides a simple and efficient solution for monitoring parking space availability. The system uses three input sensors (X0, X1, and X2), PLC ladder logic, and LED indicators to identify whether parking spaces are vacant or occupied. The internal relay M0 indicates when all three parking slots are occupied, activating the red LED (Y1), while the green LED (Y0) indicates that at least one parking space is available. The optional outputs Y2, Y3, and Y4 can display the availability of individual slots. This project demonstrates the practical application of PLC technology in automation and parking management. It can help reduce manual supervision, save drivers' time, and improve the utilization of parking spaces. In the future, the system can be enhanced by adding more parking slots, digital displays, and IoT-based monitoring features. Thus, the proposed system offers a useful foundation for developing more advanced automatic parking management systems.

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